

RAUTAVAARA YLA-LUOSTA, Finland

WMO: 027890

Lat: 63.3781N

Lon: 28.6617E

Elev: 543

StdP: 14.41

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-15.7	-26.8	1.3	-21.2	-20.5	1.8	-15.6	15.7	18.1	13.6	21.9	1.8	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	80.3	64.7	76.4	63.0	72.8	61.2	67.4	76.1	65.3	73.4	63.2	70.1	6.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	91.9	70.8	62.2	85.3	69.0	60.1	79.1	66.5	32.1	75.7	30.5	74.3	28.9	69.9	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.6	10.2	DB	-27.3	83.6	5.7	5.4	-31.4	87.5	-34.7	90.7	-37.9	93.8	-42.1	97.7	(4)
(5)			WB	-27.0	70.0	5.7	2.5	-31.0	71.8	-34.4	73.3	-37.5	74.7	-41.7	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	10.8	16.3	22.8	34.3	47.3	54.4	60.7	57.1	48.5	36.3	29.2	21.5	
	DBStd	18.61	14.83	12.94	9.94	7.18	8.33	7.23	6.58	5.98	6.07	6.54	8.20	11.81	
	HDD50	5705	1214	945	844	471	150	35	4	10	97	427	625	883	
	HDD65	10378	1679	1365	1309	921	549	326	166	254	495	891	1075	1348	
	CDD50	853	0	0	0	0	67	167	335	231	52	1	0	0	
	CDD65	51	0	0	0	0	1	8	33	9	0	0	0	0	
	CDH74	592	0	0	0	0	53	113	318	108	0	0	0	0	
	CDH80	107	0	0	0	0	3	16	68	20	0	0	0	0	
Wind	WSAvg	4.9	4.1	5.1	5.5	5.2	5.4	5.1	4.5	4.6	4.6	5.0	5.1	4.8	
Precipitation	PrecAvg	26.3	1.9	1.6	1.4	1.3	1.9	2.8	3.4	2.8	2.3	2.3	2.2	2.3	
	PrecMax	34.4	3.4	3.4	2.8	2.2	4.2	5.4	6.4	6.2	4.4	4.4	4.9	4.5	
	PrecMin	20.2	0.6	0.2	0.3	0.3	0.2	0.9	0.9	0.6	0.6	0.5	0.4	0.7	
	PrecStd	3.5	0.7	0.7	0.6	0.6	0.9	1.0	1.4	1.2	1.0	1.0	1.0	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	39.4	45.9	62.3	79.4	82.4	85.2	82.8	68.3	53.1	46.5	39.4	
		MCWB	35.7	36.3	37.7	48.8	57.9	63.5	68.4	67.2	59.9	50.6	45.7	38.5	
	2%	DB	33.3	35.5	40.6	54.8	74.4	77.6	81.4	76.6	64.1	50.5	42.8	36.9	
		MCWB	32.7	34.3	34.9	45.1	57.1	63.6	66.3	63.9	56.9	48.5	42.1	36.3	
	5%	DB	32.0	33.3	37.3	50.7	69.1	72.9	77.4	72.4	60.9	48.2	40.4	35.0	
		MCWB	31.6	32.4	33.6	41.7	54.9	60.0	64.8	61.8	55.9	46.3	39.9	34.3	
	10%	DB	30.5	32.1	35.0	46.5	63.3	68.9	73.5	68.8	58.4	45.8	38.4	33.2	
		MCWB	30.1	31.6	32.4	39.5	51.4	57.8	62.9	60.6	54.8	44.3	37.9	32.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	36.6	39.0	50.0	61.1	67.5	70.7	69.5	60.9	51.9	45.9	39.0
			MCDB	36.2	37.9	44.1	59.0	73.3	75.4	80.9	80.3	66.9	52.8	46.5	39.3
2%		WB	32.9	34.5	36.2	46.0	58.5	65.0	68.4	65.4	58.8	49.3	42.3	36.5	
		MCDB	33.1	35.6	39.0	54.9	70.4	74.5	77.3	73.4	61.9	50.2	42.7	36.8	
5%		WB	31.8	32.9	34.4	42.4	56.0	61.3	66.5	63.5	56.9	46.7	40.0	34.4	
		MCDB	32.0	33.6	36.4	48.6	66.8	70.6	75.1	69.9	60.4	47.8	40.4	34.8	
10%		WB	30.0	31.6	33.0	40.1	52.5	59.0	64.6	61.7	55.1	44.4	37.9	32.9	
		MCDB	30.5	32.1	34.6	44.9	62.1	66.6	71.8	67.4	57.8	45.6	38.3	33.2	
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.6	16.8	17.5	19.3	18.2	17.2	17.3	13.8	9.3	7.0	8.6	
		MCDBR	6.0	8.6	16.4	25.9	28.9	24.7	22.4	23.7	19.1	12.8	7.8	5.9	
		MCWBR	6.3	8.0	11.9	15.6	13.5	11.8	9.7	12.1	13.0	10.7	7.8	5.8	
	5% WB	MCDBR	5.9	8.4	12.8	21.8	25.7	21.4	20.5	19.5	16.9	11.9	7.8	6.0	
		MCWBR	6.1	7.9	9.9	13.8	13.2	11.1	10.1	10.9	12.4	10.7	7.8	6.0	
Clear-Sky Solar Irradiance	taub	0.221	0.280	0.308	0.346	0.346	0.343	0.368	0.350	0.327	0.307	0.236	0.208		
	taud	1.987	2.110	2.171	2.288	2.453	2.485	2.430	2.467	2.519	2.399	2.074	1.968		
	Ebn at Noon	121	201	247	264	276	279	267	262	246	199	114	59		
	Edh at Noon	10	21	29	33	30	30	31	28	22	16	9	5		
All-Sky Solar Radiation	RadAvg	57	233	700	1163	1499	1667	1576	1198	658	247	60	23		
	RadStd	7	54	83	133	166	103	164	134	75	42	12	3		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	-319	-344	N/A	N/A

Nomenclature: See separate page